Paper Summary

Use of Network Pharmacology and Molecular Docking to Explore the Mechanism of Action of Curcuma in the Treatment of Osteosarcoma

Osteosarcoma (OS) is one of the most common malignant bone tumors and is frequently diagnosed in adolescents. Despite standard treatment combining surgical resection with chemotherapy, OS still shows high rates of metastasis and recurrence. Extensive chemotherapy use also raises the risk of adverse reactions and drug resistance. Curcuma, a traditional Chinese medicine, has long been used as an adjunct therapy for OS because of its anticancer compounds, particularly curcumin, yet the molecular mechanism underlying its anticancer activity against OS remains unclear.

This study aimed to identify the main bioactive compounds of curcuma and clarify their mechanism of action against OS by integrating network pharmacology with molecular docking.

Anticancer compounds of curcuma were collected from relevant literature and screened for gastrointestinal absorption and drug-likeness using SwissADME, after which their potential targets were predicted through SwissTargetPrediction. OS-related targets were retrieved from GeneCards, OMIM, and PharmGKB. Overlapping targets were then analyzed through a protein–protein interaction network (STRING and Cytoscape), module clustering with MCODE, GO/KEGG enrichment analysis via DAVID, and finally validated using molecular docking with AutoDock Tool and PyMOL.

Eleven active curcuma compounds, including curcumin, curcumol, and calebin A, were identified along with 270 related targets, of which 141 overlapped with OS-related targets. Twelve hub genes were identified: AKT1, TNF, STAT3, EGFR, HSP90AA1, ESR1, EP300, ERBB2, PTGS2, MDM2, TLR4, and AR. KEGG analysis revealed involvement of the PI3K-Akt, HIF-1, and FOXO signaling pathways. Curcumin, the compound with the highest number of target connections, showed strong docking affinity toward AKT1, TNF, STAT3, and EGFR, with binding energies below −5 kJ/mol and RMSD values under 2Å, indicating stable and reliable docking results.

The strength of this study lies in its comprehensive integration of computational network pharmacology prediction with molecular docking validation, supported by multiple databases that reinforce target identification. However, the study relies entirely on in silico analysis without in vitro or in vivo experimental validation, so the predicted mechanisms remain biologically unconfirmed. In addition, differences in curcuma's effects across OS stages and the influence of dosage variation were not addressed.

This study highlights that curcuma's anticancer effect against OS operates through a multi-component, multi-target, multi-pathway mechanism, consistent with the traditional Chinese medicine principle of addressing multiple pathological aspects at once. It shows that network pharmacology is an efficient predictive tool for mapping complex herbal mechanisms before proceeding to costlier and more time-consuming experimental validation, which can help accelerate the discovery of new therapeutic targets for bone cancer.

Total: 392 words

Methods and Results Illustration

